

Project Documentation – Network Traffic Analysis (PCAP Investigation)

1. Project Overview

This documentation outlines the steps I followed during a network traffic analysis project, where I investigated a potentially malicious PCAP file. The goal was not only to detect suspicious behavior but also to extract actionable indicators of compromise (IOCs) and understand the attacker’s behavior within the network environment.

Details:

- - **Project Title:** Network Traffic Analysis of Suspicious PCAP File
- - **Date of Analysis:** 2024-09-04

2. Tools & Resources Used

To perform a proper and contextual analysis, I relied on a mix of packet analysis tools and threat intelligence platforms. Each tool had a clear purpose during my investigation:

Tool / Platform	Purpose
Wireshark	Packet inspection, filtering, and session reconstruction
VirusTotal	Checking domain/IP reputation and malware associations
ThreatFox	Crowdsourced threat intelligence data

3. Step-by-Step Analysis

Here’s a deep-dive walkthrough of my analysis workflow, from opening the PCAP file to extracting final IOCs:

3.1. Environment Setup

I started by opening the PCAP file in Wireshark and reviewing its metadata: the capture duration, number of packets, and time span. This initial step is crucial to get an overview of the scope.

➤ Initial setup in Wireshark after loading the PCAP file for analysis

The image shows the Wireshark interface with the following panes:

- Packet List:** Displays a list of 34 packets. The first packet is a DHCP Discover (344 bytes) from 0.0.0.0 to 255.255.255.255. Subsequent packets include DHCP Offer, DHCP Request, DHCP ACK, ARP, DNS, and LDAP messages.
- Packet Details:** Shows the structure of the selected packet (Packet 1: DHCP Discover). It includes fields like Ethernet II, Internet Protocol Version 4, User Datagram Protocol, and Dynamic Host Configuration Protocol (Discover).
- Packet Bytes:** Displays the raw packet data in hexadecimal and ASCII.

➤ Capture summary showing metadata including duration, packet count, and file size

The image shows the Wireshark interface with the 'Wireshark - Capture File Properties' dialog box open. The dialog displays the following information:

- File:** C:\Users\NIZAR\Desktop\2024-09-04-traffic-analysis-exercise.pcap
- Length:** 21218
- Hash (SHA256):** ff0c6d2e1686fab364f5b7471e736ad7473d5ca299f9161a4b444862e
- Hash (SHA1):** 056c3048e2e2dab5a1a510bbf4d42ef26c11a74
- Format:** Wireshark/tcpdump/- .pcap
- Encapsulation:** Ethernet
- Snapshot length:** 65535
- Time:** First packet: 2024-09-04 18:32:31, Last packet: 2024-09-04 19:32:07, Elapsed: 00:59:36
- Capture:** Hardware: Unknown, OS: Unknown, Application: Unknown
- Interfaces:** Interface: Unknown, Dropped packets: Unknown, Capture filter: Unknown, Link type: Ethernet, Packet size limit (snapshot): 65535 bytes
- Statistics:** Measurement: Captured (5091), Displayed (5091 (100.0%)), Marked (—). Time span: 3576.160, Average pps: 1.4, Average packet size: 402, Bytes: 2048179, Average bytes/s: 572, Average bits/s: 4581.

3.2. Traffic Filtering

Using display filters, I narrowed down the traffic to the host of interest: 172.17.0.99. Filters like `dns`, `http.request`, `tls.handshake`, and `ip.addr == 172.17.0.99` allowed me to isolate relevant sessions and discard noise from unrelated flows.

➤ Filtered the traffic to display only DNS communications involving the internal host 172.17.0.99

The image shows a Wireshark packet capture window with the display filter set to `ip.addr==172.17.0.99 && dns`. The packet list pane displays a series of DNS messages between 172.17.0.99 and 172.17.0.17. The packet details pane shows the structure of a DNS Standard query response, including the question section with the query name `www.msftncsl.com` and the answer section with the IP address `23.220.251.149`. The packet bytes pane shows the raw data of the DNS response, including the header, question, and answer sections.

No.	Time	Source	Destination	Protocol	Length	Info
7	0.010136	172.17.0.99	172.17.0.17	DNS	126	Standard query 0xbfb6 SRV _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.bepositive.com
9	0.010335	172.17.0.17	172.17.0.99	DNS	192	Standard query response 0xbfb6 SRV _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.bepositive.com SRV 0 100 389 win-ctl9xbqy19.bepositive.com A 172.17.0.17
12	0.011106	172.17.0.99	172.17.0.17	DNS	90	Standard query 0xc088 SOA DESKTOP-RN09AT.bepositive.com
13	0.011106	172.17.0.99	172.17.0.17	DNS	90	Standard query 0xc3329 A win-ctl9xbqy19.bepositive.com
14	0.011286	172.17.0.17	172.17.0.99	DNS	169	Standard query response 0xc088 SOA DESKTOP-RN09AT.bepositive.com SOA win-ctl9xbqy19.bepositive.com A 172.17.0.17
15	0.011286	172.17.0.17	172.17.0.99	DNS	106	Standard query response 0xc3329 A win-ctl9xbqy19.bepositive.com A 172.17.0.17
17	0.011888	172.17.0.99	172.17.0.17	DNS	156	Dynamic update 0xc3c44 SOA.bepositive.com CHAHE AAAA A 172.17.0.99
19	0.013951	172.17.0.17	172.17.0.99	DNS	156	Dynamic update response 0xc3c44 SOA.bepositive.com CHAHE AAAA A 172.17.0.99
23	0.113075	172.17.0.99	172.17.0.17	DNS	126	Standard query 0xbac7 SRV _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.bepositive.com
25	0.113199	172.17.0.17	172.17.0.99	DNS	192	Standard query response 0xbac7 SRV _ldap._tcp.Default-First-Site-Name._sites.dc._msdcs.bepositive.com SRV 0 100 389 win-ctl9xbqy19.bepositive.com A 172.17.0.17
39	0.508024	172.17.0.99	172.17.0.17	DNS	79	Standard query 0xcab7 A upad.bepositive.com
40	0.508024	172.17.0.99	172.17.0.17	DNS	79	Standard query 0xc113 A upad.bepositive.com
41	0.508954	172.17.0.17	172.17.0.99	DNS	156	Standard query response 0xcab7 No such name A upad.bepositive.com SOA win-ctl9xbqy19.bepositive.com
42	0.508954	172.17.0.17	172.17.0.99	DNS	156	Standard query response 0xc113 No such name A upad.bepositive.com SOA win-ctl9xbqy19.bepositive.com
47	0.516987	172.17.0.99	172.17.0.17	DNS	83	Standard query 0x2282 A www.msftconnecttest.com
48	0.575239	172.17.0.17	172.17.0.99	DNS	227	Standard query response 0x2282 A www.msftconnecttest.com CHAHE ncsi-geo.trafficmanager.net CHAHE www.msftncsl.com.edgesuite.net CHAHE a1961.g2.akamai.net A 23.220.251.149 A 23.220...
67	1.074000	172.17.0.99	172.17.0.17	DNS	83	Standard query 0xf4de A www.msftconnecttest.com
68	1.675541	172.17.0.17	172.17.0.99	DNS	227	Standard query response 0xf4de A www.msftconnecttest.com CHAHE ncsi-geo.trafficmanager.net CHAHE www.msftncsl.com.edgesuite.net CHAHE a1961.g2.akamai.net A 23.220.251.158 A 23.220...
105	6.614502	172.17.0.99	172.17.0.17	DNS	76	Standard query 0xc585 A dns.msftncsl.com
106	6.655105	172.17.0.17	172.17.0.99	DNS	92	Standard query response 0xc585 A dns.msftncsl.com A 131.107.255.255
109	18.518099	172.17.0.99	172.17.0.17	DNS	79	Standard query 0x7717 A upad.bepositive.com
110	18.518714	172.17.0.99	172.17.0.17	DNS	79	Standard query 0xeb91 A upad.bepositive.com
111	18.518828	172.17.0.17	172.17.0.99	DNS	156	Standard query response 0x7717 No such name A upad.bepositive.com SOA win-ctl9xbqy19.bepositive.com
112	18.518709	172.17.0.17	172.17.0.99	DNS	156	Standard query response 0xeb91 No such name A upad.bepositive.com SOA win-ctl9xbqy19.bepositive.com
152	41.729652	172.17.0.99	172.17.0.17	DNS	83	Standard query 0x5325 A pti.store.microsoft.com
153	41.788827	172.17.0.17	172.17.0.99	DNS	156	Standard query response 0x5325 No such name A pti.store.microsoft.com SOA ns1-39.azure-dns.com
213	78.534338	172.17.0.99	172.17.0.17	DNS	79	Standard query 0xbf69 A upad.bepositive.com
214	78.534338	172.17.0.99	172.17.0.17	DNS	79	Standard query 0xbf69 A upad.bepositive.com
215	78.534338	172.17.0.17	172.17.0.99	DNS	156	Standard query response 0xbf69 No such name A upad.bepositive.com SOA win-ctl9xbqy19.bepositive.com
216	78.534401	172.17.0.17	172.17.0.99	DNS	156	Standard query response 0xbf69 No such name A upad.bepositive.com SOA win-ctl9xbqy19.bepositive.com
243	184.325074	172.17.0.99	172.17.0.17	DNS	89	Standard query 0xc3c2 A v10.events.data.microsoft.com
244	184.379593	172.17.0.17	172.17.0.99	DNS	226	Standard query response 0xc3c2 A v10.events.data.microsoft.com CHAHE win-global-asimov-leafs-events-data.trafficmanager.net CHAHE onedscolpreu15.eastus.cloudapp.azure.com A 20.42...
498	124.514020	172.17.0.99	172.17.0.17	DNS	72	Standard query 0x762d A www.bing.com
499	124.574405	172.17.0.17	172.17.0.99	DNS	337	Standard query response 0x762d A www.bing.com CHAHE www.bing.com.trafficmanager.net CHAHE www.bing.com.edgekey.net CHAHE e86303.dscx.akamaiedge.net A 23.221.24.58 A 23.221.24.5...
528	124.732930	172.17.0.99	172.17.0.17	DNS	74	Standard query 0x5cfa A assets.msn.com
531	124.788037	172.17.0.17	172.17.0.99	DNS	180	Standard query response 0x5cfa A assets.msn.com CHAHE assets.msn.com.edgesuite.net CHAHE e28578.d.akamaiedge.net A 23.45.119.144 A 23.45.119.143
565	124.905559	172.17.0.99	172.17.0.17	DNS	71	Standard query 0xbab8 A arc.msn.com
575	125.010317	172.17.0.17	172.17.0.99	DNS	186	Standard query response 0xbab8 A arc.msn.com CHAHE arc.trafficmanager.net CHAHE iris-de-prod-azsc-v2-eas2.eastus2.cloudapp.azure.com A 20.96.153.111
599	125.333830	172.17.0.99	172.17.0.17	DNS	71	Standard query 0x215e A www.msn.com
602	125.378807	172.17.0.17	172.17.0.99	DNS	146	Standard query response 0x215e A www.msn.com CHAHE www.msn.com.a-0003.a-msedge.net CHAHE a-0003.a-msedge.net A 204.79.197.203
668	125.660740	172.17.0.99	172.17.0.17	DNS	86	Standard query 0x0398 A officeclient.microsoft.com
805	125.879711	172.17.0.17	172.17.0.99	DNS	235	Standard query response 0x0398 A officeclient.microsoft.com CHAHE config.officeapps.live.com CHAHE prod.configsvcl1.live.com.akadns.net CHAHE us.configsvcl1.live.com.akadns.net CHAHE...
885	126.412816	172.17.0.99	172.17.0.17	DNS	83	Standard query 0x5344 A odc.officeapps.live.com
886	126.554620	172.17.0.17	172.17.0.99	DNS	233	Standard query response 0x5344 A odc.officeapps.live.com CHAHE prod.odcml1.live.com.akadns.net CHAHE us2.odcml1.live.com.akadns.net CHAHE vus-azsc-000.odc.officeapps.live.com CHAHE...

Frame 575: 186 bytes on wire (1488 bits), 186 bytes captured (1488 bits) on 0
Ethernet II, Src: Dell_S01ba:fd (00:23:ae:50:ba:fd), Dst: Intel_b6:8d:c4 (18:3d:a2:b6:8d:c4)
Internet Protocol Version 4, Src: 172.17.0.17, Dst: 172.17.0.99
User Datagram Protocol, Src Port: 53, Dst Port: 5366
Domain Name System (response)

Packets: 5091 · Displayed: 173 (3.4%)

➤ Protocol-level filters used to isolate HTTP, and TLS communications for deeper inspection

The screenshot shows a Wireshark capture of a TLS handshake. The packet list on the left shows a series of messages between 172.17.0.99 and 10.10.31.115. The selected packet is a TLSv1.2 Client Hello (232 bytes). The packet details pane on the right shows the structure of the Client Hello, including the TLS version (3.1), cipher suites, and compression methods. The packet bytes pane on the right shows the raw hex and ASCII data.

The screenshot shows a Wireshark capture of an HTTP GET request. The packet list on the left shows a series of messages between 172.17.0.99 and 172.17.0.99. The selected packet is an HTTP GET request (222 bytes). The packet details pane on the right shows the structure of the GET request, including the method (GET), the URI (/assets/), and the content type (text/html). The packet bytes pane on the right shows the raw hex and ASCII data.

3.3. Domain & DNS Analysis

I examined all domain queries made by the internal host. One domain stood out:

www[.]bellantonicioccolato[.]it, which hinted at potentially pirated or malicious content.

DNS query observed for suspicious domain **www[.]bellantonicciocolato[.]it**

The image shows a Wireshark packet capture of a network traffic analysis. The top pane displays a list of packets, with packet 1279 highlighted in red. The middle pane shows the details of the selected packet, which is a DNS query (Standard Query Query) for the domain **www.bellantonicciocolato.it**. The bottom pane shows the raw packet data in hexadecimal and ASCII.

No.	Time	Source	Destination	Protocol	Length	Info
1153	128.996254	172.17.0.99	40.126.28.12	TLSv1.3	424	Change Cipher Spec, Client Hello (SNI=login.microsoftonline.com)
1158	129.054959	172.17.0.99	23.40.145.142	TLSv1.2	326	Client Hello (SNI=oneclient.sfx.ms)
1159	129.060966	40.126.28.12	172.17.0.99	TLSv1.3	1430	Server Hello
1171	129.094118	23.40.145.142	172.17.0.99	TLSv1.2	1430	Server Hello
1176	129.094577	23.40.145.142	172.17.0.99	TLSv1.2	1430	Certificate
1179	129.113772	23.40.145.142	172.17.0.99	TLSv1.2	1399	Certificate Status, Server Key Exchange, Server Hello Done
1181	129.122153	172.17.0.99	23.40.145.142	TLSv1.2	180	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
1189	129.161674	23.40.145.142	172.17.0.99	TLSv1.2	312	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
1203	129.308485	172.17.0.99	23.195.212.189	TLSv1.2	343	Client Hello (SNI=ssodaddy.com)
1205	129.344418	23.195.212.189	172.17.0.99	TLSv1.2	1430	Server Hello
1210	129.345011	23.195.212.189	172.17.0.99	TLSv1.2	1430	Certificate
1212	129.372802	23.195.212.189	172.17.0.99	TLSv1.2	1430	Certificate Status
1213	129.372889	23.195.212.189	172.17.0.99	TLSv1.2	78	Server Key Exchange, Server Hello Done
1215	129.380955	172.17.0.99	23.195.212.189	TLSv1.2	180	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
1217	129.427485	23.195.212.189	172.17.0.99	TLSv1.2	328	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
1249	145.434001	172.17.0.99	13.107.246.57	TLSv1.3	366	Client Hello (SNI=inputuggestions.msdcn.microsoft.com)
1251	145.468288	13.107.246.57	172.17.0.99	TLSv1.3	153	Hello Retry Request, Change Cipher Spec
1253	145.468938	172.17.0.99	13.107.246.57	TLSv1.3	405	Change Cipher Spec, Client Hello (SNI=inputuggestions.msdcn.microsoft.com)
1254	145.508272	13.107.246.57	172.17.0.99	TLSv1.3	1430	Server Hello, Application Data
1279	153.718782	172.17.0.99	79.124.78.197	TLSv1.3	1430	Server Hello, Change Cipher Spec, Application Data
1533	155.079782	172.17.0.99	40.126.28.12	TLSv1.3	353	Client Hello (SNI=login.microsoftonline.com)
1547	155.163404	40.126.28.12	172.17.0.99	TLSv1.3	153	Hello Retry Request, Change Cipher Spec
1549	155.167316	172.17.0.99	40.126.28.12	TLSv1.3	424	Change Cipher Spec, Client Hello (SNI=login.microsoftonline.com)
1597	155.238303	40.126.28.12	172.17.0.99	TLSv1.3	1430	Server Hello
1635	155.460855	172.17.0.99	23.195.212.189	TLSv1.2	343	Client Hello (SNI=ssodaddy.com)
1637	155.518230	23.195.212.189	172.17.0.99	TLSv1.2	1430	Server Hello
1640	155.518575	23.195.212.189	172.17.0.99	TLSv1.2	1430	Certificate
1645	155.554025	23.195.212.189	172.17.0.99	TLSv1.2	1430	Certificate Status
1646	155.554025	23.195.212.189	172.17.0.99	TLSv1.2	78	Server Key Exchange, Server Hello Done
1648	155.567892	172.17.0.99	23.195.212.189	TLSv1.2	180	Client Key Exchange, Change Cipher Spec, Encrypted Handshake Message
1653	155.675091	23.195.212.189	172.17.0.99	TLSv1.2	328	New Session Ticket, Change Cipher Spec, Encrypted Handshake Message
1704	158.130216	172.17.0.99	46.254.34.201	TLSv1.3	332	Client Hello (SNI=www.bellantonicciocolato.it)
1706	158.307947	46.254.34.201	172.17.0.99	TLSv1.3	1430	Server Hello, Change Cipher Spec, Application Data
2400	246.075132	172.17.0.99	204.79.197.203	TLSv1.3	479	Client Hello (SNI=windows.msn.com)
2402	246.075420	172.17.0.99	204.79.197.203	TLSv1.3	447	Client Hello (SNI=windows.msn.com)
2405	246.145589	204.79.197.203	172.17.0.99	TLSv1.3	153	Hello Retry Request, Change Cipher Spec
2406	246.147446	172.17.0.99	204.79.197.203	TLSv1.3	721	Change Cipher Spec, Client Hello (SNI=windows.msn.com)
2409	246.156285	204.79.197.203	172.17.0.99	TLSv1.3	153	Hello Retry Request, Change Cipher Spec
2410	246.157443	172.17.0.99	204.79.197.203	TLSv1.3	680	Change Cipher Spec, Client Hello (SNI=windows.msn.com)
2412	246.226064	204.79.197.203	172.17.0.99	TLSv1.3	1430	Server Hello
2419	246.227079	204.79.197.203	172.17.0.99	TLSv1.3	1430	Server Hello
2445	246.517954	172.17.0.99	204.79.197.203	TLSv1.3	443	Client Hello (SNI=www.msn.com)
2447	246.592151	204.79.197.203	172.17.0.99	TLSv1.3	153	Hello Retry Request, Change Cipher Spec

The domain was flagged as **suspicious** by only one antivirus engine on VirusTotal, while all other engines reported it as clean

The image shows the VirusTotal scan results for the domain **www.bellantonicciocolato.it**. The scan shows a community score of 0/95, indicating that no engines have flagged the domain as malicious. The scan details show that the domain is associated with a Delphi loader, which is a sophisticated malware payload. The security vendors' analysis shows that only one engine, Forcepoint ThreatSeeker, has flagged the domain as suspicious, while all other engines have reported it as clean.

Security vendor	Result
Forcepoint ThreatSeeker	Suspicious
Acronis	Clean
ALL Labs (MONITORAPP)	Clean
Antiy-AVL	Clean
BitDefender	Clean
Certego	Clean
Abusix	Clean
ADMINUS Labs	Clean
AlienVault	Clean
benkow.cc	Clean
Blueliv	Clean
Chong Lua Dao	Clean

3.4. IP Address Correlation

The internal host 172.17.0.99 communicated with several external IPs. One notable IP, **79.124.78.197**, was cross-checked against **VirusTotal** and **ThreatFox** and was found to be associated with malware activity, including potential command-and-control communication. This suggests the host was likely compromised and attempting to communicate with malicious infrastructure.

➤ Repeated communication observed with suspicious external IP : 79.124.78.197

The image displays a Wireshark packet capture analysis of a network traffic file named '2024-09-04-traffic-analysis-exercise.pcap'. The interface shows the 'Packet List' pane on the left, the 'Packet Details' pane in the middle, and the 'Packet Bytes' pane on the right.

The 'Packet List' pane shows a list of captured packets. The first packet (No. 52) is a GET request to 'connecttest.txt' from source IP 172.17.0.99 to destination IP 23.220.251.149. Subsequent packets (No. 1668 to 4691) are POST requests to '/foots.php' from source IP 172.17.0.99 to destination IP 79.124.78.197. These packets are highlighted with a red box, indicating the repeated communication with the suspicious external IP.

The 'Packet Details' pane shows the details of the selected packet (No. 52), which is a GET request to 'connecttest.txt' from source IP 172.17.0.99 to destination IP 23.220.251.149. The details include the Ethernet II header, Internet Protocol Version 4 header, Transmission Control Protocol header, and Hypertext Transfer Protocol header.

The 'Packet Bytes' pane shows the raw bytes of the selected packet, including the Ethernet II header, Internet Protocol Version 4 header, Transmission Control Protocol header, and Hypertext Transfer Protocol header.

➤ VirusTotal confirms the IP address is **malicious**

The screenshot shows the VirusTotal web interface for the IP address 79.124.78.197. The page has a dark theme. At the top, a notification states '3/95 security vendors flagged this IP address as malicious'. Below this, a circular 'Community Score' of 3/95 is displayed. The IP address is listed as 79.124.78.197 (79.124.78.0/24) with AS 208370 (DigiSys Ltd.). The 'DETECTION' tab is active, showing a table of security vendors' analysis. A banner at the top of the table asks 'Do you want to automate checks?'. The table lists 20 vendors, with BitDefender, G-Data, and CyRadar marked as 'Malicious' or 'Malware', while the others are 'Clean'.

Security vendors' analysis		Do you want to automate checks?	
BitDefender	Malware	CyRadar	Malicious
G-Data	Malware	Abusix	Clean
Acronis	Clean	ADMINUSLabs	Clean
AllLabs (MONITORAPP)	Clean	AlienVault	Clean
alphaMountain.ai	Clean	Antiy-AVL	Clean
benkow.cc	Clean	Blueliv	Clean
Certego	Clean	Chong Lua Dao	Clean
CINS Army	Clean	CMC Threat Intelligence	Clean
CRDF	Clean	Criminal IP	Clean
Cyble	Clean	desenmascara.me	Clean

➤ Threat Intelligence lookup confirms the IP is linked to known Malware (**Koi Loader**)

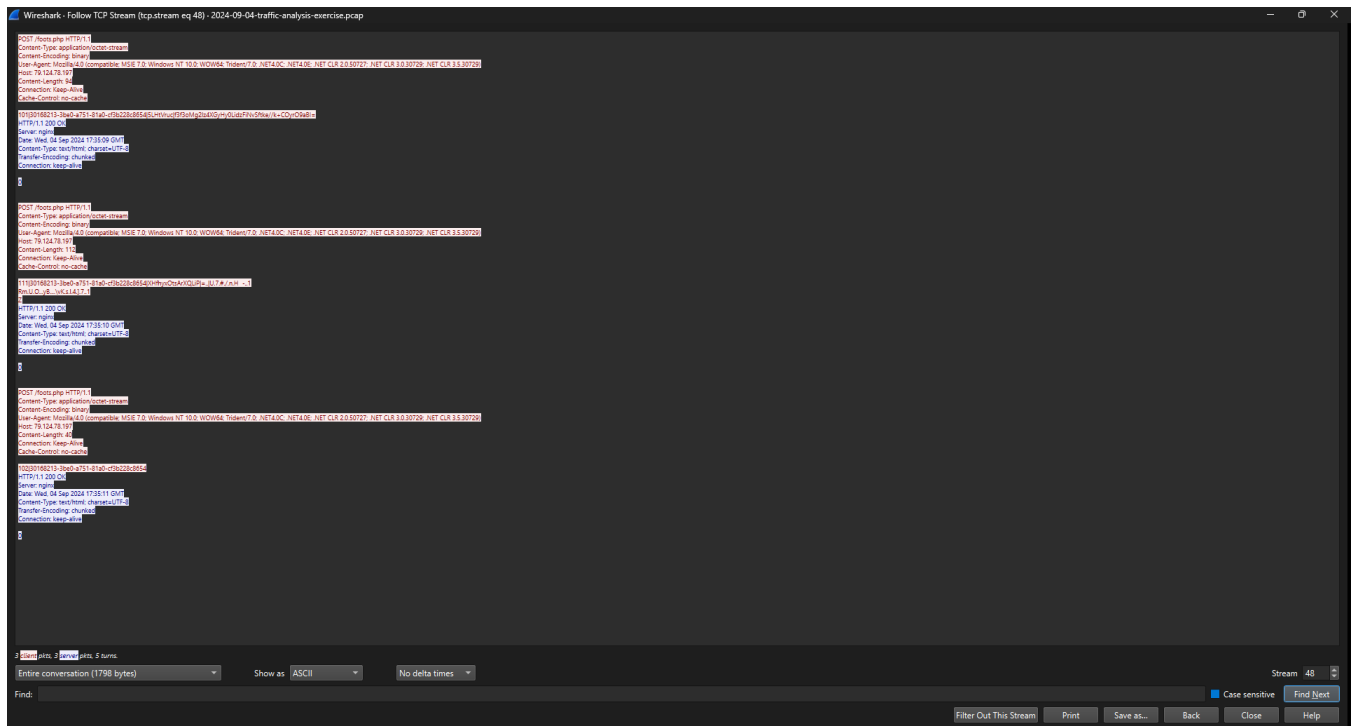
The screenshot shows the ThreatFox web interface for the IOC http://79.124.78.197/foots.php. The page has a dark theme. A yellow banner at the top says 'Authenticate for API access'. Below this, a message states 'This IOC expired' with a note that the IOC is old and has expired on 2025-09-17. The main content area displays a table of IOC details. The 'IOC ID' is 1319448. The 'IOC' is http://79.124.78.197/foots.php. The 'IOC Type' is url. The 'Threat Type' is botnet_cc. The 'Malware' is Koi Loader. The 'Confidence Level' is high (100%). The 'ASN' is AS201133 Verdina. The 'Country' is BG. The 'First seen' is 2024-09-01 18:47:50 UTC. The 'Last seen' is never. The 'UUID' is b013f88c-6892-11ef-ac38-42010aa4000a. The 'Reporter' is AndreGiorda. The 'Reward' is 5 credits from ThreatFox. The 'Tags' are KoiLoader. The 'Reference' is https://tria.ge/240901-w33m7aygje.

IOC ID:	1319448
IOC:	http://79.124.78.197/foots.php
IOC Type:	url
Threat Type:	botnet_cc
Malware:	Koi Loader
Confidence Level:	Confidence level is high (100%)
ASN:	AS201133 Verdina
Country:	BG
First seen:	2024-09-01 18:47:50 UTC
Last seen:	never
UUID:	b013f88c-6892-11ef-ac38-42010aa4000a
Reporter:	AndreGiorda
Reward:	5 credits from ThreatFox
Tags:	KoiLoader
Reference:	https://tria.ge/240901-w33m7aygje

1.1. HTTP / POST Traffic Review

I identified a POST request sent over HTTP (not HTTPS) to the IP mentioned below. The URL path and parameters suggested possible data exfiltration or beaconsing behavior.

➤ **Encrypted HTTP POST request made to suspicious IP, indicating potential data exfiltration.**



1.2. Victim Profiling

From internal metadata and traffic patterns, I was able to extract partial victim profiling data such as hostname, MAC address, internal IP, and hints of the system username. This helped assess the impact scope and identify the potentially compromised asset

The image shows a Wireshark packet capture analysis of network traffic. The filter applied is `ip.addr==172.17.0.99 && kerberos`. The packet list shows several packets, with packet 303 highlighted. The packet details pane for packet 303 shows the structure of the KRB5 AS-REP message, including the session key and the target principal name.

No.	Time	Source	Destination	Protocol	Length	Info
31	0.114391	172.17.0.99	172.17.0.17	LDAP	668	bindRequest(7) "croot/" sasl
33	0.115167	172.17.0.17	172.17.0.99	LDAP	264	bindResponse(7) success
120	26.898403	172.17.0.99	172.17.0.17	SWB2	814	Session Setup Request
122	26.898454	172.17.0.17	172.17.0.99	SWB2	314	Session Setup Response
293	123.984999	172.17.0.99	172.17.0.17	KRB5	283	AS-REQ
303	123.992901	172.17.0.17	172.17.0.99	KRB5	363	AS-REP
312	123.994211	172.17.0.99	172.17.0.17	KRB5	231	TGS-REQ
317	123.996874	172.17.0.17	172.17.0.99	KRB5	255	TGS-REP
343	124.162229	172.17.0.99	172.17.0.17	KRB5	388	TGS-REQ
346	124.163392	172.17.0.99	172.17.0.17	KRB5	404	TGS-REQ
350	124.165399	172.17.0.17	172.17.0.99	KRB5	377	TGS-REP
354	124.165656	172.17.0.17	172.17.0.99	KRB5	361	TGS-REP
358	124.166801	172.17.0.99	172.17.0.17	DCERPC	751	Bind: call_id: 2, Fragment: Single, 3 context items: DRSUAPI V4.0 (32bit NDR), DRSUAPI V4.0 (64bit NDR), DRSUAPI V4.0 (64bit NDR)
368	124.166599	172.17.0.99	172.17.0.17	KRB5	181	TGS-REQ
370	124.167683	172.17.0.17	172.17.0.99	DCERPC	338	Bind_ack: call_id: 2, Fragment: Single, max_xmit: 5840 max_recv: 5840, 3 results: Provider rejection, Acceptance, Negotiate ACK
371	124.167889	172.17.0.99	172.17.0.17	DCERPC	274	Alter_context: call_id: 2, Fragment: Single, 1 context items: DRSUAPI V4.0 (64bit NDR)
373	124.168024	172.17.0.17	172.17.0.99	KRB5	210	TGS-REP
381	124.168713	172.17.0.99	172.17.0.17	SWB2	736	Session Setup Request
390	124.171149	172.17.0.17	172.17.0.99	SWB2	314	Session Setup Response
412	124.173967	172.17.0.99	172.17.0.17	LDAP	656	bindRequest(27) "croot/" sasl
419	124.175920	172.17.0.17	172.17.0.99	LDAP	264	bindResponse(27) success
443	124.183514	172.17.0.99	172.17.0.17	LDAP	642	bindRequest(31) "croot/" sasl
452	124.184258	172.17.0.17	172.17.0.99	LDAP	264	bindResponse(31) success
480	124.256169	172.17.0.99	172.17.0.17	KRB5	373	TGS-REQ
483	124.257488	172.17.0.17	172.17.0.99	KRB5	331	TGS-REP
490	124.258123	172.17.0.99	172.17.0.17	SWB2	691	Session Setup Request
492	124.258854	172.17.0.17	172.17.0.99	SWB2	314	Session Setup Response
1068	128.743056	172.17.0.99	172.17.0.17	DCERPC	751	Bind: call_id: 2, Fragment: Single, 3 context items: DRSUAPI V4.0 (32bit NDR), DRSUAPI V4.0 (64bit NDR), DRSUAPI V4.0 (64bit NDR)
1070	128.744569	172.17.0.17	172.17.0.99	DCERPC	338	Bind_ack: call_id: 2, Fragment: Single, max_xmit: 5840 max_recv: 5840, 3 results: Provider rejection, Acceptance, Negotiate ACK
1071	128.744779	172.17.0.99	172.17.0.17	DCERPC	274	Alter_context: call_id: 2, Fragment: Single, 1 context items: DRSUAPI V4.0 (64bit NDR)
1095	128.754125	172.17.0.99	172.17.0.17	LDAP	642	bindRequest(3) "croot/" sasl
1097	128.755698	172.17.0.17	172.17.0.99	LDAP	264	bindResponse(3) success
1108	128.759733	172.17.0.99	172.17.0.17	LDAP	642	bindRequest(9) "croot/" sasl
1110	128.760527	172.17.0.17	172.17.0.99	LDAP	264	bindResponse(9) success
1435	154.968820	172.17.0.99	172.17.0.17	DCERPC	751	Bind: call_id: 2, Fragment: Single, 3 context items: DRSUAPI V4.0 (32bit NDR), DRSUAPI V4.0 (64bit NDR), DRSUAPI V4.0 (64bit NDR)
1437	154.970230	172.17.0.17	172.17.0.99	DCERPC	338	Bind_ack: call_id: 2, Fragment: Single, max_xmit: 5840 max_recv: 5840, 3 results: Provider rejection, Acceptance, Negotiate ACK
1438	154.970619	172.17.0.99	172.17.0.17	DCERPC	274	Alter_context: call_id: 2, Fragment: Single, 1 context items: DRSUAPI V4.0 (64bit NDR)
1462	154.980905	172.17.0.99	172.17.0.17	LDAP	642	bindRequest(3) "croot/" sasl
1465	154.982451	172.17.0.17	172.17.0.99	LDAP	264	bindResponse(3) success
1476	154.987135	172.17.0.99	172.17.0.17	LDAP	642	bindRequest(9) "croot/" sasl
1478	154.988681	172.17.0.17	172.17.0.99	LDAP	264	bindResponse(9) success
3265	184.955303	172.17.0.99	172.17.0.17	DCERPC	789	Bind: call_id: 2, Fragment: Single, 3 context items: DRSUAPI V4.0 (32bit NDR), DRSUAPI V4.0 (64bit NDR), DRSUAPI V4.0 (64bit NDR)

Packet 303 details:

- Frame 480, 373 bytes on wire (2984 bits), 373 bytes captured (2984 bits)
- Ethernet II, Src: Intel_b8:bd:c4 (18:3d:a2:b8:bd:c4), Dst: Dell_50:b1:af:d (00:23:ae:50:b1:af:d)
- Internet Protocol Version 4, Src: 172.17.0.99, Dst: 172.17.0.17
- Transmission Control Protocol, Src Port: 49786, Dst Port: 88, Seq: 1461, Ack: 1, Len: 319
- [2 Reassembled TCP Segments (1779 bytes): #479(1468), #480(319)]
- Kerberos

Packet 303 details (hex):

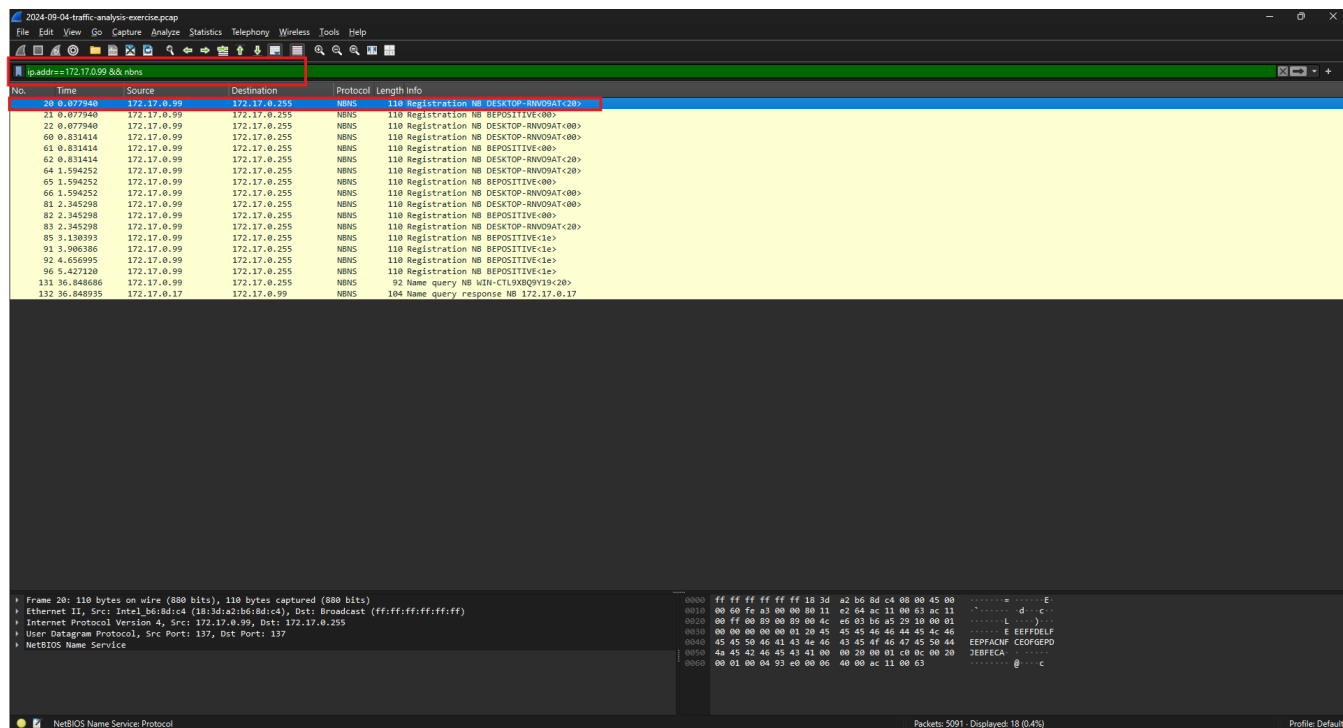
```
0000 00 23 ae 50 b1 af d 18 3d a2 b6 bd c4 00 00 45 00  #P...E
0010 01 67 16 8c 40 00 00 06 8a 6e ac 11 00 03 ac 11  g...n...c
0020 00 11 c2 7a 00 58 61 0e 0f 88 59 4a 13 76 50 18  zXa...YJ.vP
0030 10 04 e2 b0 00 00 48 bd 1d b1 6f 87 02 02 b5 de  ....H...O
0040 79 47 ee dc 02 ed c0 f0 c2 7e 03 3a ef 5a 9d 30  yG.....:O
0050 15 a1 04 02 02 00 a7 a2 0d 04 0b 30 09 a0 07 03  .......O
0060 05 00 40 00 00 04 82 01 0b 30 82 01 07 a0 07  @.....O
0070 03 05 00 40 01 00 00 a2 10 1b 0e 42 45 50 4f 53  @.....DEPOS
0080 49 54 49 56 45 2e 43 4f 4d a3 22 30 20 a0 03 02  ITIVE.CO M"O
```

[illegible]

The image displays a Wireshark packet capture of a Lightweight Directory Access Protocol (LDAP) session. The top pane shows the packet list with 3018 packets. The middle pane shows the packet details for packet 3018, highlighting the LDAP search request and response. The bottom pane shows the raw packet data in hexadecimal and ASCII.

Packet List (Top Pane):

No.	Time	Source	Destination	Protocol	Length	Info
2297	184.983602	172.17.0.99	172.17.0.17	LDAP	238	SASL GSS-API Privacy: payload (120 bytes)
2298	184.983606	172.17.0.17	172.17.0.99	LDAP	215	SASL GSS-API Privacy: payload (97 bytes)
2299	184.984138	172.17.0.99	172.17.0.17	LDAP	232	SASL GSS-API Privacy: payload (114 bytes)
2300	184.984612	172.17.0.17	172.17.0.99	LDAP	268	SASL GSS-API Privacy: payload (150 bytes)
2305	184.989650	172.17.0.17	172.17.0.17	LDAP	680	bindRequest(9) "ROOT" sasl
2307	184.991238	172.17.0.17	172.17.0.99	LDAP	264	bindResponse(9) success
2308	184.991617	172.17.0.99	172.17.0.17	LDAP	172	SASL GSS-API Privacy: payload (54 bytes)
2311	184.992516	172.17.0.17	172.17.0.99	LDAP	255	SASL GSS-API Privacy: payload (1807 bytes)
2312	184.992601	172.17.0.99	172.17.0.17	LDAP	305	SASL GSS-API Privacy: payload (187 bytes)
2314	184.993106	172.17.0.17	172.17.0.99	LDAP	330	SASL GSS-API Privacy: payload (212 bytes)
2315	184.993478	172.17.0.99	172.17.0.17	LDAP	129	SASL GSS-API Privacy: payload (11 bytes)
2317	184.993554	172.17.0.99	172.17.0.17	LDAP	129	SASL GSS-API Privacy: payload (11 bytes)
2318	319.549171	172.17.0.99	172.17.0.17	LDAP	404	searchRequest(1) "ROOT" baseObject
3000	319.549508	172.17.0.17	172.17.0.99	LDAP	1375	searchResult(1) "ROOT" searchDone(1) success [2 results]
3001	319.554614	172.17.0.99	172.17.0.17	LDAP	588	bindRequest(3) "ROOT" sasl
3002	319.554614	172.17.0.99	172.17.0.17	LDAP	264	bindResponse(3) success
3018	319.557295	172.17.0.99	172.17.0.17	LDAP	212	SASL GSS-API Integrity: searchRequest(4) "CN=Andrew Fletcher,CN=Users,DC=positive,DC=com" baseObject
3019	319.557912	172.17.0.17	172.17.0.99	LDAP	236	SASL GSS-API Integrity: searchResultEntry(4) "CN=Andrew Fletcher,CN=Users,DC=positive,DC=com" searchDone(4) success [2 results]
3020	319.558016	172.17.0.99	172.17.0.17	LDAP	97	SASL GSS-API Integrity: searchResultEntry(5) "CN=Andrew Fletcher,CN=Users,DC=positive,DC=com" searchDone(5) success [1 result]
3020	319.558686	172.17.0.17	172.17.0.99	LDAP	588	bindRequest(8) "ROOT" sasl
3030	319.568381	172.17.0.17	172.17.0.99	LDAP	264	bindResponse(8) success
3031	319.568686	172.17.0.99	172.17.0.17	LDAP	212	SASL GSS-API Integrity: searchRequest(9) "CN=Andrew Fletcher,CN=Users,DC=positive,DC=com" baseObject
3033	319.569151	172.17.0.17	172.17.0.99	LDAP	236	SASL GSS-API Integrity: searchResultEntry(9) "CN=Andrew Fletcher,CN=Users,DC=positive,DC=com" searchDone(9) success [1 result]
3033	319.569378	172.17.0.99	172.17.0.17	LDAP	97	SASL GSS-API Integrity: unbindRequest(10)
3034	425.436705	172.17.0.99	172.17.0.17	LDAP	404	searchRequest(1) "ROOT" baseObject
3035	425.437411	172.17.0.17	172.17.0.99	LDAP	1375	searchResult(1) "ROOT" searchDone(1) success [2 results]
3101	425.438596	172.17.0.99	172.17.0.17	LDAP	588	bindRequest(3) "ROOT" sasl
3101	425.440519	172.17.0.17	172.17.0.99	LDAP	264	bindResponse(3) success
3104	425.441603	172.17.0.99	172.17.0.17	LDAP	212	SASL GSS-API Integrity: searchRequest(4) "CN=Andrew Fletcher,CN=Users,DC=positive,DC=com" baseObject
3105	425.442084	172.17.0.17	172.17.0.99	LDAP	236	SASL GSS-API Integrity: searchResultEntry(4) "CN=Andrew Fletcher,CN=Users,DC=positive,DC=com" searchDone(4) success [2 results]
3108	425.442332	172.17.0.99	172.17.0.17	LDAP	97	SASL GSS-API Integrity: unbindRequest(5)
3112	425.448548	172.17.0.99	172.17.0.17	LDAP	588	bindRequest(8) "ROOT" sasl
3114	425.450070	172.17.0.17	172.17.0.99	LDAP	264	bindResponse(8) success
3115	425.450297	172.17.0.99	172.17.0.17	LDAP	212	SASL GSS-API Integrity: searchRequest(9) "CN=Andrew Fletcher,CN=Users,DC=positive,DC=com" baseObject
3116	425.450760	172.17.0.17	172.17.0.99	LDAP	236	SASL GSS-API Integrity: searchResultEntry(9) "CN=Andrew Fletcher,CN=Users,DC=positive,DC=com" searchDone(9) success [1 result]
3117	425.450858	172.17.0.99	172.17.0.17	LDAP	97	SASL GSS-API Integrity: unbindRequest(10)
3226	726.026753	172.17.0.				



2. Key Observations

- The internal host 172.17.0.99 initiated multiple SMB connections with unusual payloads, including buffer overflow attempts, indicating possible exploitation activity.
- Suspicious HTTP POST requests to the external IP **79.124.78.197** suggest potential command-and-control communication or data exfiltration.
- Multiple communications targeted high-risk services such as **SMB** and **Kerberos**, which could enable privilege escalation or lateral movement.
- The IOC 79.124.78.197 was cross-checked against **VirusTotal** and **ThreatFox**, showing associations with malware infrastructure, confirming malicious activity.

3. IOC Table

Type	Value	Source / Notes
IP	79.124.78.197	Malicious IP associated with Koi Loader Malware
Domain	www[.]bellantonicioccolato[.]it	Flagged as suspicious on VirusTotal
URL	Hxxp[://]79[.]124[.]78[.]197/foots[.]php	Observed in HTTP POST

7. What I Learned

This project reinforced important Blue Team and DFIR skills, including:

- ⇒ Identifying and analyzing suspicious SMB and Kerberos activity within internal traffic.
- ⇒ Isolating and investigating malicious external communications using Wireshark and filtering techniques.
- ⇒ Cross-referencing IOCs such as IPs and URLs with VirusTotal, ThreatFox, and other threat intelligence platforms.
- ⇒ Recognizing potential malware behavior, including C2 communication and data exfiltration attempts.
- ⇒ Documenting and presenting findings in a clear, structured manner suitable for incident response and reporting.